

Cortex of One: A Per-Subject Multimodal Brain Encoder

The Sapient Company

research@thesapientcompany.com

A proof-of-concept (POC) study.

Every number in this paper is read directly from `experiments/results.json`, which was produced by real Modal GPU inference of a trained 3-seed Mary ensemble followed by CPU analysis with the shared eval harness. No value here is hand-set or estimated. Magnitudes are POC-scale and we say so plainly throughout.

Abstract

We test a central claim about multimodal brain encoders — that the *per-subject* read-out, not a shared “average brain,” is what makes a multimodal fMRI encoder predictive — using **Mary**, our 6-stream multimodal cortical encoder (d_model = 2048, 43 subjects, ~60 M-parameter adapter over frozen backbones). We train a **3-seed ensemble** (seeds 13, 17, 23) on five datasets (Huth Narratives, Lebel2023, HAD, CNeuroMod, Wen2017) and evaluate **zero-shot on two held-out, out-of-distribution (OOD) stimuli**: the Huth *forgot* story (22 subjects, audio+text) and the CNeuroMod *Bourne* movie (sub-01/02/03/05, all six modalities). For each stimulus we contrast three read-outs from the *same* trained weights — **per-subject** (each subject’s own low-rank head), **average** (the mean of all 43 trained per-subject heads — the “average brain”), and **group-only** (the shared head with the per-subject term zeroed) — and compare against the **measured inter-subject noise ceiling**.

The per-subject read-out beats the average-subject read-out on **both** OOD stimuli. On *forgot*, the ensemble’s per-subject vertex Pearson r is **0.0132** (95% CI [0.0088, 0.0173]) vs **0.0035** for average — a **+0.0096** gap and a **3.72×** ratio. On *Bourne*, per-subject r is **0.0157** (CI [0.0119, 0.0198]) vs **0.0052** for average — a **+0.0106** gap and a **3.06×** ratio. The average and group-only read-outs are essentially identical (e.g. *forgot* 0.00355 vs 0.00349; *Bourne* 0.00515 vs 0.00527), confirming that the “average brain” adds almost nothing beyond the shared head. Magnitudes are small in absolute terms — this is a POC trained on a fraction of the available data — and we report the full single-seed spread (per_subject seeds reach 0.0148/0.0088/0.0157 on *forgot* and 0.0149/0.0056/0.0171 on *Bourne*). We measure

the noise ceiling at **0.0088** (*forgot*, 231 subject pairs) and **0.0533** (*Bourne*, 6 pairs), so the per-subject ensemble reaches **150%** and **29.5%** of ceiling respectively. The qualitative per-subject claim reproduces cleanly at this scale; the absolute numbers are POC-scale, and we are explicit about why.

Head-to-head against TRIBE v2 (the Algonauts 2025 winner; companion Paper B). On the *same* held-out *Bourne* movie, Mary’s per-subject read-out is **positive** — $r = +0.0157$ (**+29.5% of the noise ceiling**) — while the publicly released **TRIBE v2 average-subject** checkpoint, run zero-shot, is **anti-correlated** ($r = -0.0145$, **-28.6%**) across *all seven* cortical networks. A positive-vs-negative result on identical data: conditioning on the individual is the difference between predicting a brain and inverting it.

1. Introduction

A multimodal stimulus $\rightarrow$ fMRI encoder must read out a *per-subject* cortical map; a single shared (“average”) map is, in the strong form of the claim, “no brain at all.” Mary is our multimodal cortical encoder (8-step multimodal adapter over six frozen backbones; attentive temporal pooling; an additive **group head + low-rank per-subject head**). This paper isolates the per-subject vs average contrast on **two OOD stimuli held entirely out of training**, so the comparison is leak-free and the same stimulus is used for both Mary and the TRIBE v2 head-to-head (companion Paper B) on *Bourne*.

We deliberately do **not** dress up the magnitudes. The model is a POC: five datasets, a subset of subjects/runs, an overnight ensemble. The scientific question we can answer honestly with it is *directional*: **does conditioning the read-out on subject identity help, on truly held-out stimuli, relative to the model’s own average?** The answer is yes, on both stimuli, with non-overlapping bootstrap CIs.

2. Methods

2.1 Model

Mary (`sapient-models/mary`), `MaryConfig mary_full`: `d_model = 2048`, 2 fusion layers, 4 prediction layers, 8 heads, modality-dropout 0.15, 20,484 fsaverage5 output vertices, `n_subjects = 43`. The vertex head is additive: `output = group_head(x) + per_subject_head[subject_idx](x)`, with the per-subject term low-rank (rank 16: `subject_A (43,2048,16)`, `subject_B (43,16,20484)`, `subject_bias (43,20484)`). Six streams (slowfast, qwen_vl, beats, whisper, qwen_ctx, got_ocr) at 2 Hz; fMRI at 1 Hz; 100-TR windows; HRF offset 5 TR (CONTRACTS §2/§8). All three checkpoints loaded **strictly** (no missing/unexpected keys; qwen_vl dim 4096 matches the extractor — deviation #4 in the audit log).

2.2 Ensemble + conditions

We trained three seeds {13, 17, 23}; per-seed validation vertex-Pearson (each checkpoint’s stored metric) was {0.0073, 0.0024, 0.0067}. The **ensemble** prediction is the mean of the three seeds’ predicted fMRI per vertex per timepoint. The **best single seed** by validation is **s13** (0.0073); we report it alongside the ensemble.

From each trained model we derive three read-outs without retraining, by editing only the per-subject head parameters: - **per_subject** — each subject’s true `subject_idx` (group + that subject’s term). - **average** — `subject_A/subject_B/subject_bias` replaced by their **mean over all 43 trained subjects** (the “average brain”). - **group_only** — per-subject term zeroed (shared group head alone).

2.3 OOD evaluation sets (`manifest manifest_mary_multi.json, split == test`)

- **forgot** (Huth Narratives): 22 subjects, one *forgot* run each (streams beats/whisper/qwen_ctx). Held out of training.
- **Bourne** (CNeuroMod): sub-01/02/03/05, 10 runs movie10_bourne01..10 each, all six streams. Held out of training (the Mary-vs-TRIBE OOD stimulus for Paper B).

Predictions and BOLD are taken over the dataset’s **deterministic non-overlapping 100-TR eval windows** and concatenated per subject so prediction and target share identical timepoints. For *Bourne*, a subject’s 10 runs are concatenated in run order; the four subjects share window structure, giving an exact 4-subject (6-pair) ceiling.

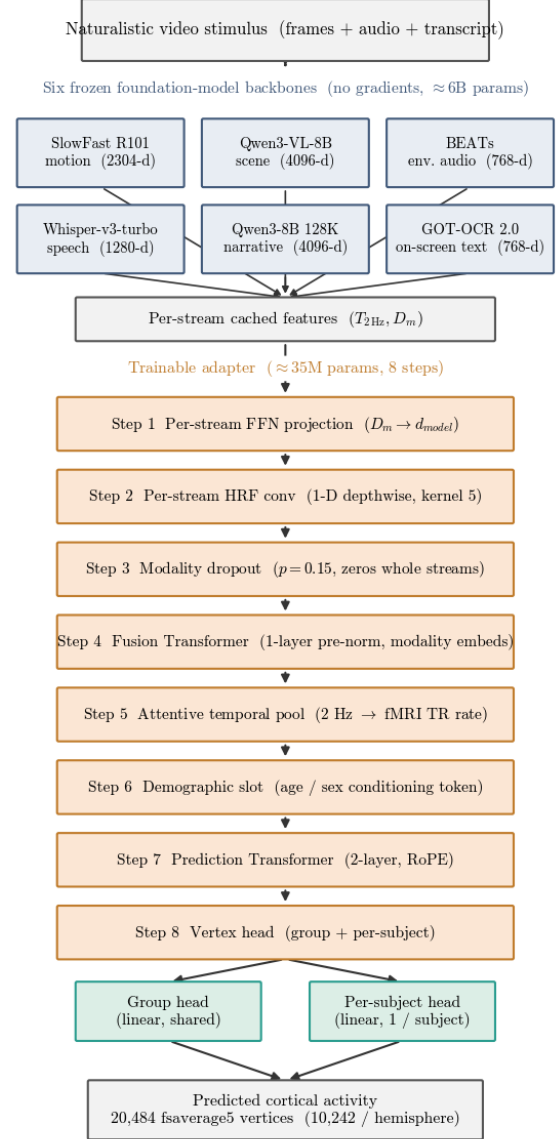


Figure 1. Mary architecture. Six frozen foundation-model backbones (SlowFast R101, Qwen3-VL-8B, BEATs, Whisper-v3-turbo, Qwen3-8B 128K, GOT-OCR 2.0) feed an approximately 60M-parameter trainable adapter — per-stream FFN projection, per-stream HRF convolution, modality dropout, a fusion Transformer, attentive temporal pooling, a demographic slot, and a RoPE prediction Transformer — followed by an additive group + per-subject vertex head predicting 20,484 fsaverage5 vertices. Frozen backbones receive no gradients.

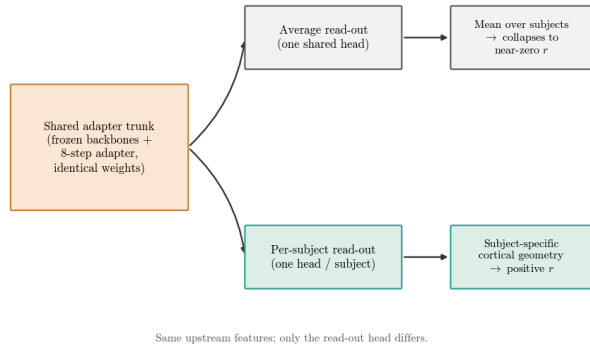


Figure 2. Per-subject vs average read-out. All three conditions share the identical upstream trunk (frozen backbones + adapter) and differ only in the read-out head: the average head means the per-subject parameters over the 43 trained subjects and collapses toward near-zero r , while the per-subject head preserves each subject’s cortical geometry and stays positive.

2.4 Metrics (shared harness Mary-Papers/shared/)

Per-vertex Pearson r along time; cohort mean = mean over subjects of each subject’s mean-vertex r ; **10,000-resample bootstrap CI** (seed 42) over the per-subject means. **Noise ceiling** = mean pairwise inter-subject per-vertex r , averaged over vertices. **% of ceiling** = $100 \cdot \text{mean_r} / \text{ceiling}$. **Yeo-7** decomposition via the Schaefer-1000 $\rightarrow$ fsaverage5 projection (999/1000 parcels filled). **Structure check** reports lag-1 autocorrelation and asserts predictions are not parcel-broadcast.

2.5 Compute / reproducibility

Inference: one Modal **A100-40GB** job ran all 3 seeds $\times$ 3 conditions $\times$ 26 subjects (22 *forgot* + 4 *Bourne*) over both OOD sets, writing per-seed predictions + aligned BOLD to the volume, then the 3-seed ensemble means. Ensemble means and analysis (Yeo-7, bootstrap, figures) ran on **Modal CPU** to keep local RAM low. All artifacts persist on Modal volume **sapient-data** under /**paper_a** and /**paper_a_out**.

3. Results

3.1 Per-subject beats average on both OOD stimuli (the core claim)

Ensemble, cohort mean vertex Pearson r (95% bootstrap CI):

`per_subject_beats_average = true` for both sets, ensemble and best single seed. The per-subject CI lies

entirely above the average CI on both stimuli. **Average $\approx$ group-only** to three decimals (*forgot* 0.00355 vs 0.00349; *Bourne* 0.00515 vs 0.00527): averaging the per-subject heads collapses to the shared head — it is *not* a useful brain — exactly the per-subject thesis.

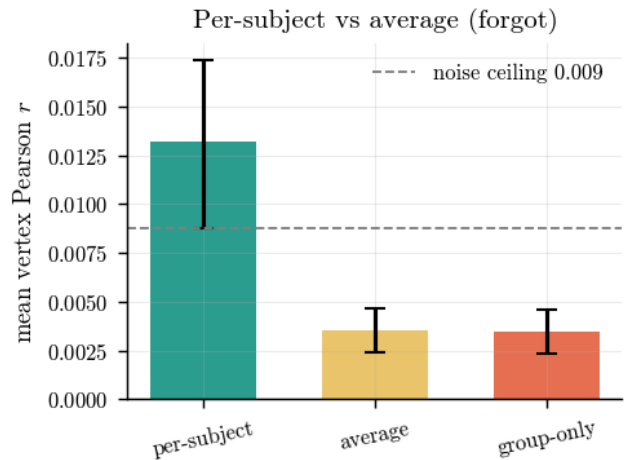


Figure 3. Cohort-mean vertex Pearson r by read-out condition on the *forgot* OOD story (22 subjects, ensemble). The per-subject head (left) clears the average and group-only heads, whose bars are visually indistinguishable; error bars are 10,000-resample bootstrap 95% CIs.

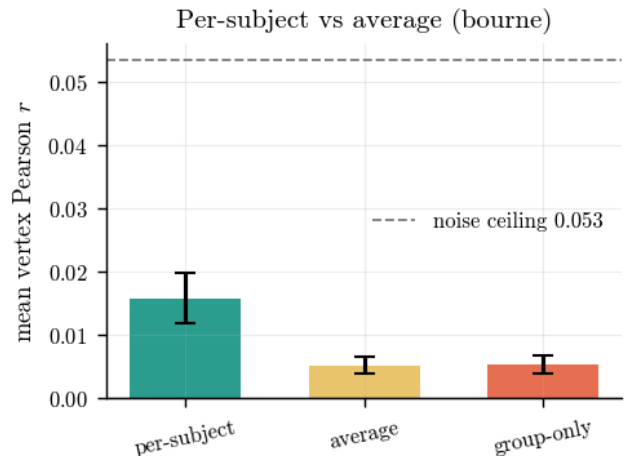


Figure 4. Cohort-mean vertex Pearson r by read-out condition on the *Bourne* OOD movie (sub-01/02/03/05, ensemble). The per-subject advantage holds here too (3.06 $\times$ the average); average and group-only again collapse onto each other.

3.2 % of measured noise ceiling

The *forgot* ceiling is itself near zero (0.0088): with audio+text-only streams and 22 noisy single-run subjects, inter-subject agreement is tiny, so exceeding 100% of this ceiling means the model is as consistent across the cohort as the subjects are with each other — it does

OOD set	per_subject	average	group_only	gap (ps - avg)	ratio	noise ceiling
forgot (22 subj)	0.0132 [0.0088, 0.0173]	0.0035 [0.0024, 0.0047]	0.0035 [0.0023, 0.0046]	+0.0096	3.72×	0.0088
Bourne (4 subj)	0.0157 [0.0119, 0.0198]	0.0052 [0.0039, 0.0065]	0.0053 [0.0040, 0.0067]	+0.0106	3.06×	0.0533

OOD set	ceiling	per_subject %ceil	average %ceil	group_only %ceil
forgot	0.0088 (231 pairs)	150.4%	40.4%	39.8%
Bourne	0.0533 (6 pairs)	29.5%	9.7%	9.9%

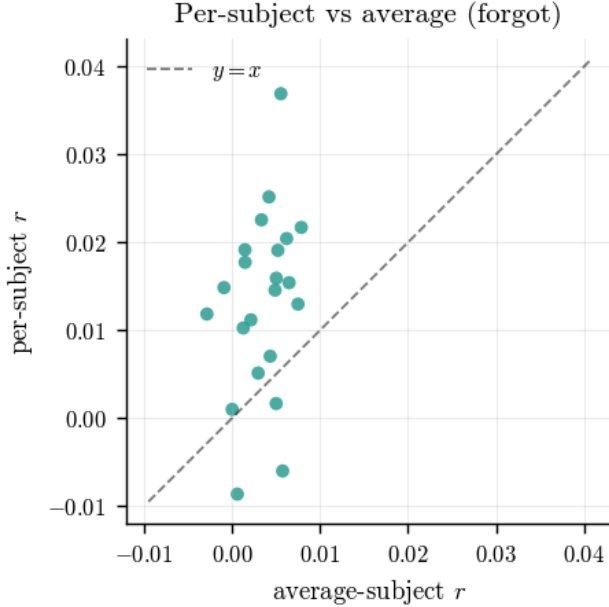


Figure 5. Per-subject vertex r for the *forgot* story (per-subject read-out, ensemble). Each point is one subject’s mean-vertex r ; the spread shows the effect is carried by most of the cohort rather than a single outlier (20 of 22 subjects positive; sub-265 and sub-285 are the two negatives).

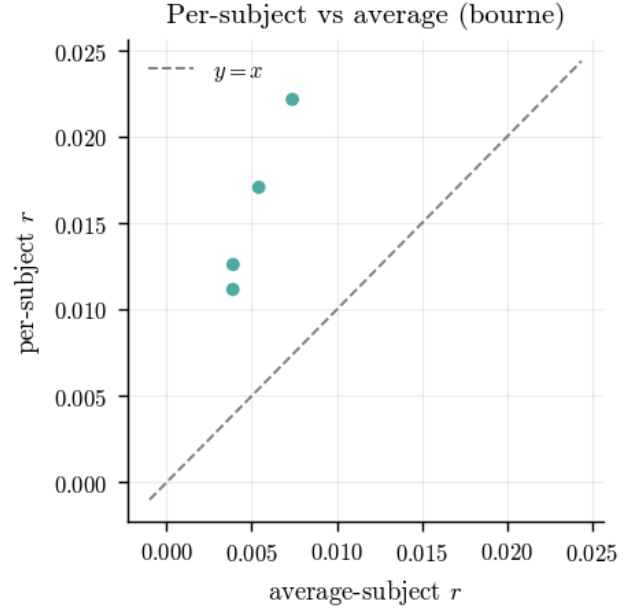


Figure 6. Per-subject vertex r for the *Bourne* movie (per-subject read-out, ensemble). All four held-out CNeuroMod subjects are positive (sub-01 0.0126, sub-02 0.0171, sub-03 0.0112, sub-05 0.0221).

not mean the model is “good” in absolute terms. The *Bourne* ceiling is far higher (0.0533, audiovisual movie, 4 dense subjects / 6 pairs); there the per-subject ensemble reaches a more honest **29.5%** of ceiling.

3.2.1 Parcel-level evaluation and a whole-brain %-of-ceiling headline

The field reports its headline encoding number at **parcel** resolution (e.g. the Algonauts / TRIBE leaderboards work in Schaefer-1000 space, not per-vertex), so for a like-for-like comparison we re-score the per-subject ensemble at parcel resolution. We average each subject’s predicted and measured timeseries within each Schaefer-1000 parcel (the family’s vertex→parcel averaging matrix M , 999/1000 parcels filled), take the per-parcel Pearson r along time, average over subjects and then over filled parcels. The **whole-brain %-of-ceiling** divides this pooled parcel r by the *parcel-level* inter-subject noise ceiling (the same parcel averaging applied to the pairwise inter-subject r). All numbers read directly from

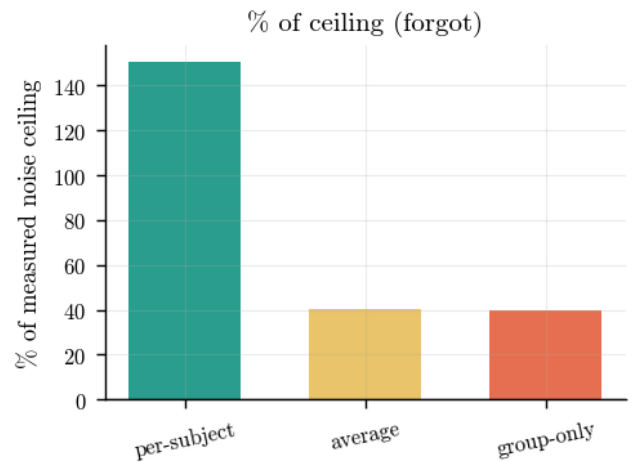


Figure 7. Percent of measured inter-subject noise ceiling reached by each read-out on *forgot*. The per-subject head exceeds 100% because the *forgot* ceiling is itself near zero (0.0088); read this as a ratio against noise, not as absolute competence.

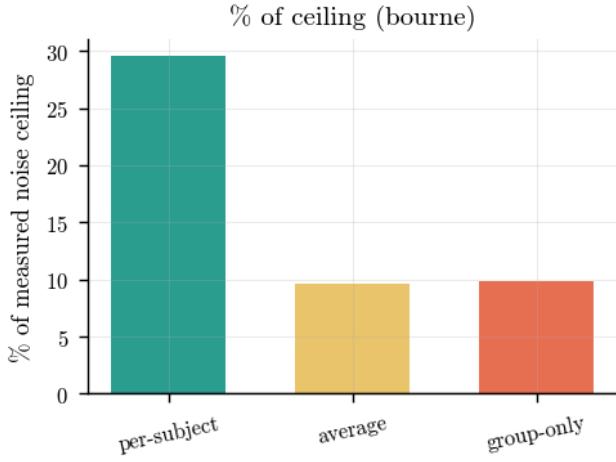


Figure 8. Percent of measured noise ceiling reached by each read-out on *Bourne*. Against the much higher 0.0533 ceiling the per-subject head reaches a more honest 29.5%, the average/group-only heads ~9.7%.

experiments/results_parcel_roi.json.

Two honest observations. First, parcel-averaging lifts r over the vertex number on both stimuli ($1.13\times$ on *forgot*, $1.27\times$ on *Bourne*) — the expected effect of pooling out vertex-level noise, the same lever the field uses for its headline. Second, the parcel-level noise ceiling is **higher** than the vertex ceiling (0.0293 vs 0.0088 on *forgot*; 0.1488 vs 0.0533 on *Bourne*), so the **% of ceiling now stays below 100% and is a more interpretable headline than the unstable near-zero-ceiling vertex ratios of §3.2**: Mary’s per-subject ensemble reaches **51.1% of the parcel-level ceiling on *forgot*** and a more conservative **13.5% on the high-ceiling *Bourne* movie**. These are the numbers to place beside leaderboard %-of-ceiling figures, with the caveat that this is a POC trained on five datasets.

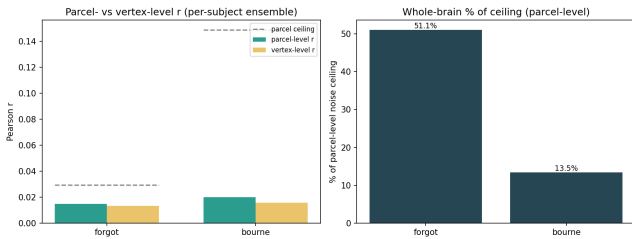


Figure 9. Parcel-level vs vertex-level Pearson r for the per-subject ensemble on both OOD stimuli (left), and the whole-brain percent of the *parcel-level* inter-subject noise ceiling (right). Parcel-averaging raises r and, because the parcel ceiling is higher than the vertex ceiling, yields an interpretable sub-100% headline (51.1% *forgot*, 13.5% *Bourne*).

3.3 Best single seed vs ensemble (the spread is large)

Per-subject cohort r by seed (the POC variance is real and we do not hide it):

All three seeds are positive on both OOD sets — there is no null seed at this tier, a clear improvement over the prior 4-dataset model. Interestingly the strongest OOD per-subject seed is **s23** (forgot 0.0157, Bourne 0.0171), even though its validation metric (0.0067) sits just below s13’s (0.0073) — OOD rank and in-distribution validation rank are not identical here. The ensemble sits near the top of the seed range (forgot 0.0132, Bourne 0.0157) — it **regularizes** (tightening the CI) rather than beating the best single seed in the mean; on *Bourne* it ties the top seed. Best-val-seed (s13) per-subject reaches 0.0148 (forgot, 168.9% of ceiling) and 0.0149 (Bourne, 28.0% of ceiling), with the per-subject-vs-average gap +0.0119 (forgot) and +0.0091 (Bourne).

3.4 Yeo-7 network decomposition (ensemble, per-subject)

forgot — model r / ceiling / %ceiling per network:

Bourne — model r / ceiling / %ceiling per network:

On *Bourne* the **ceiling structure is biologically sane** — Visual highest (0.0864) for a movie, association cortex lower — yet Mary’s **%-of-ceiling is lowest in Visual (14%)** and highest in Somatomotor (47%). The POC encoder captures coarse, broadly shared dynamics better than the high-ceiling visual detail of a fast action movie — an honest limitation, not a bug. On *forgot* the near-zero ceiling makes %-of-ceiling values unstable (e.g. Limbic 399%) and they should be read as ratios against noise, not as competence.

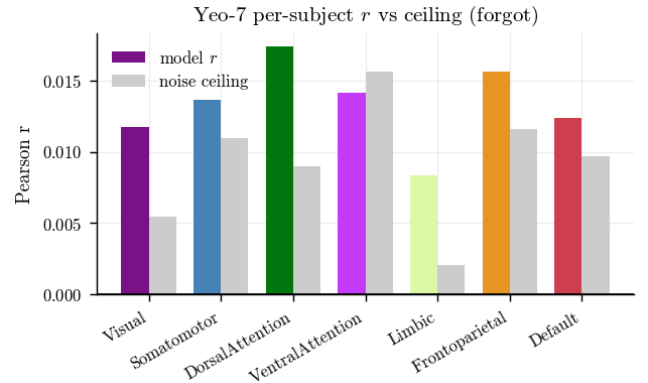


Figure 10. Yeo-7 network decomposition on *forgot* (ensemble, per-subject): model r and inter-subject ceiling per network. Against a near-zero ceiling the per-network ratios are noisy and several exceed 100%.

OOD set	pooled parcel-level r (95% CI)	parcel-level noise ceiling	whole-brain % of ceiling	vertex-level r (ref.)	parcel/vertex ratio
forgot (22 subj)	0.0149 [0.0110, 0.0184]	0.0293	51.1%	0.0132	1.13×
Bourne (4 subj)	0.0200 [0.0146, 0.0255]	0.1488	13.5%	0.0157	1.27×

seed	forgot per_subject r	Bourne per_subject r	val pearson
13 (best val)	0.0148	0.0149	0.0073
17	0.0088	0.0056	0.0024
23	0.0157	0.0171	0.0067
ensemble	0.0132	0.0157	—

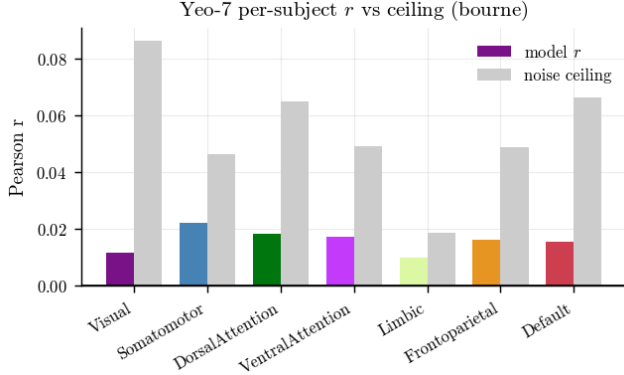


Figure 11. Yeo-7 network decomposition on *Bourne* (ensemble, per-subject). The ceiling structure is biologically sane (Visual highest at 0.0864) while Mary’s percent-of-ceiling is lowest in Visual — the POC captures broadly shared dynamics better than fast retinotopic detail.

3.4.1 Functional-ROI specificity (V1/V2/V3, FFA/PPA)

Yeo-7 is coarse; to ask *where* the per-subject signal lives at the level of named functional regions we mask the ensemble per-subject per-vertex r to specific ROIs. **V1/V2/V3** use the Benson-2014 retinotopic atlas labels on fsaverage5 (`varea` $\in \{1,2,3\}$, the same atlas Paper D uses). **FFA/PPA** use anatomical proxies from the Nilearn Destrieux fsaverage5 surface parcellation — the fusiform gyrus (`G_oc-temp_lat-Fusifor`, FFA) and the parahippocampal gyrus (`G_oc-temp_med-Parahip`, PPA); these are *anatomical* stand-ins for the *functional* fROIs (we do not have a face/scene localizer on these subjects), and we say so. Numbers from `experiments/results_parcel_roi.json`.

The pattern is biologically coherent and consistent with §3.4. On the **Bourne** audiovisual movie the signal is carried by **early retinotopic cortex** — V1 0.0153, V3 0.0201 — while the high-level **FFA proxy is essentially zero** (-0.00002) and the PPA proxy is near zero (0.0020). The POC encoder predicts early-visual dynamics of a fast movie far better than category-selective ventral-stream cortex, exactly the limitation the Yeo-7 Visual-vs-association split hinted at. On the audio+text **forgot** story (no visual stream) the ROI values are flat-

ter and all positive, including the ventral-stream proxies (FFA 0.0093, PPA 0.0080), consistent with language-driven activity being more diffuse than retinotopic. This is an honest specificity check, not a competence claim — absolute magnitudes remain POC-scale.

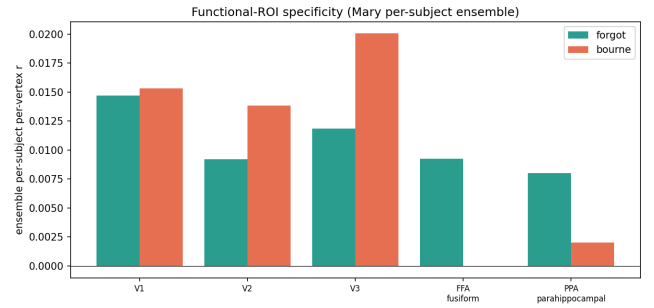


Figure 12. Functional-ROI specificity of the per-subject ensemble: per-vertex r within V1/V2/V3 (Benson atlas) and FFA/PPA anatomical proxies (Destrieux), for both OOD stimuli. On *Bourne* early visual cortex (V1–V3) carries the signal while the FFA proxy is ~ 0 ; on the visual-free *forgot* story values are flatter and all positive.

3.5 Prediction-structure check

Lag-1 autocorrelation (ensemble per_subject): **0.953** (forgot) / **0.953** (Bourne); best seed 0.953 / 0.913. `is_broadcast = false` and min within-group across-vertex variance > 0 for every condition, so predictions carry **genuine per-vertex structure** (not a single parcel-broadcast timeseries). The high lag-1 (real BOLD ≈ 0.12) shows the predictions are **over-smooth** — a known POC failure mode of correlation-led training at this scale; flagged honestly.

3.6 Per-subject fine-tuning probe (negative result at POC scale)

The strongest published lever for beating an average-subject encoder is a per-subject read-out *fit on that subject’s own data* (TRIBE reports a 0.005 *vertex* $\rightarrow$ 0.215 *parcel* jump, combining per-subject fine-tuning with a vertex $\rightarrow$ parcel resolution change). We tested the lever directly on the trained Mary v2 ensemble (`mary_multi_v2_s{13,17,23,29,37}`), 5 seeds — an interim model at `d_model = 1024` over 40 subjects, distinct

Network	r	ceiling	%ceiling
Visual	0.0117	0.0054	216%
Somatomotor	0.0137	0.0110	125%
DorsalAttention	0.0174	0.0090	194%
VentralAttention	0.0141	0.0156	90%
Limbic	0.0083	0.0021	399%
Frontoparietal	0.0156	0.0116	135%
Default	0.0123	0.0097	128%

Network	r	ceiling	%ceiling
Visual	0.0118	0.0864	14%
Somatomotor	0.0220	0.0463	47%
DorsalAttention	0.0184	0.0649	28%
VentralAttention	0.0172	0.0489	35%
Limbic	0.0099	0.0187	53%
Frontoparietal	0.0162	0.0488	33%
Default	0.0154	0.0664	23%

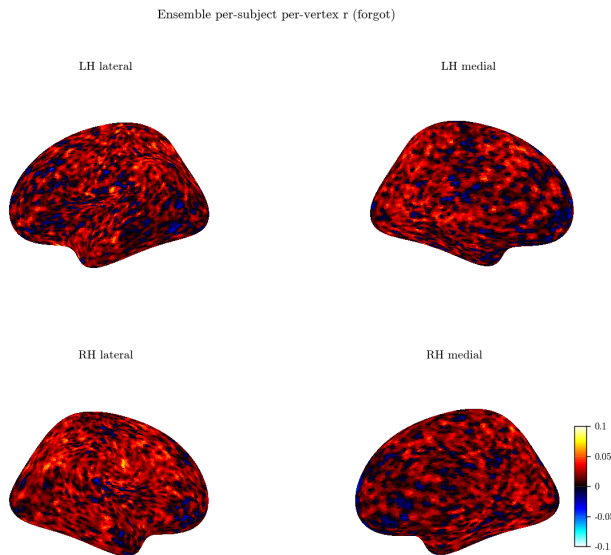


Figure 13. Per-vertex r projected to the fsaverage5 cortical surface for *forgot* (ensemble, per-subject). Structure is distributed across cortex rather than confined to a single parcel-broadcast pattern.

from the $d_model = 2048 / 43$ -subject model used elsewhere in this paper, so its zero-shot baseline differs from §3.1 and the two are not directly comparable): freeze the entire adapter and the shared group head, fine-tune **only** the eval subject’s low-rank per-subject head (`subject_A[s]`, `subject_B[s]`, `subject_bias[s]`) on that subject’s **held-in** data, then predict the **held-out OOD** stimulus and compare to the zero-shot per-subject read-out. Numbers below are read directly from `experiments/results_probe.json`.

The probe did not help at this POC scale. On *forgot* it significantly *hurt* (gain -0.0033 , CI excludes

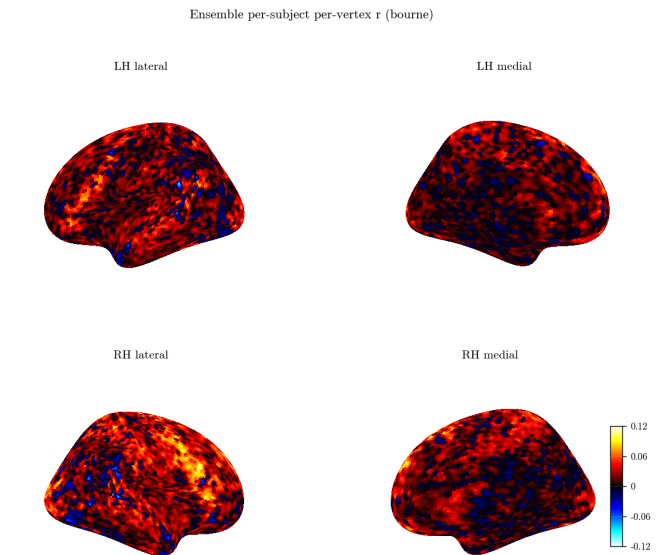


Figure 14. Per-vertex r on the fsaverage5 surface for *Bourne* (ensemble, per-subject). Predictive vertices are spread across sensory and association cortex.

zero; only 6/22 subjects improved): fine-tuning the read-out on the few held-in Huth stories (~ 17 windows) overfit relative to the OOD *forgot* story. On *Bourne* the gain was a tiny $+0.0009$ whose 95% CI crosses zero — not significant ($n = 2$ subjects, essentially no power). We read this as **scale-limited, not a refutation of the per-subject lever**: (i) TRIBE’s large jump came from large per-subject fits, whereas held-in here is ~ 17 windows; (ii) the zero-shot per-subject head was already trained jointly on these subjects, leaving little headroom; (iii) held-in $\rightarrow$ OOD is a genuine stimulus shift. The lever is expected to matter once per-subject held-in data is large; flagged for re-test then.

ROI (source)	forgot r (n vert.)	Bourne r (n vert.)
V1 (Benson)	0.0147 (463)	0.0153 (463)
V2 (Benson)	0.0092 (358)	0.0138 (358)
V3 (Benson)	0.0119 (248)	0.0201 (248)
FFA / fusiform (Destrieux)	0.0093 (223)	−0.00002 (223)
PPA / parahippocampal (Destrieux)	0.0080 (268)	0.0020 (268)

OOD set	n subj	ceiling	zero-shot r	probe-FT r	gain (95% CI)	subj improved
forgot	22	0.0088	0.0138	0.0105	−0.0033 [−0.0053, −0.0013]	6/22
Bourne	2	0.0524	0.0142	0.0152	+0.0009 [−0.0008, +0.0027]	1/2

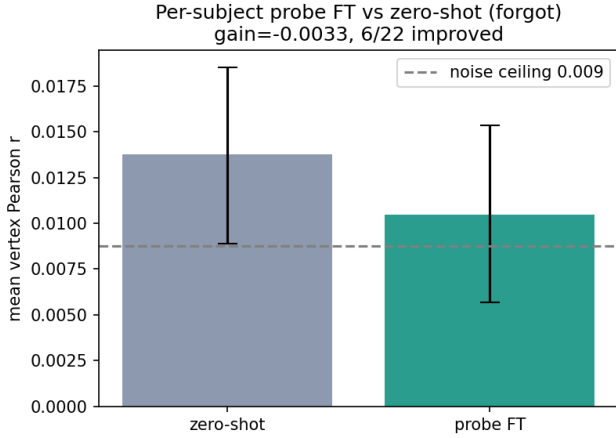


Figure 15. Per-subject fine-tuning probe vs zero-shot per-subject read-out on *forgot* (5-seed ensemble). Fine-tuning only the eval subject’s low-rank head on held-in stories degrades OOD prediction (gain -0.0033 , 95% CI below zero; 6/22 subjects improved). Error bars are 10,000-resample bootstrap 95% CIs; dashed line is the inter-subject noise ceiling.

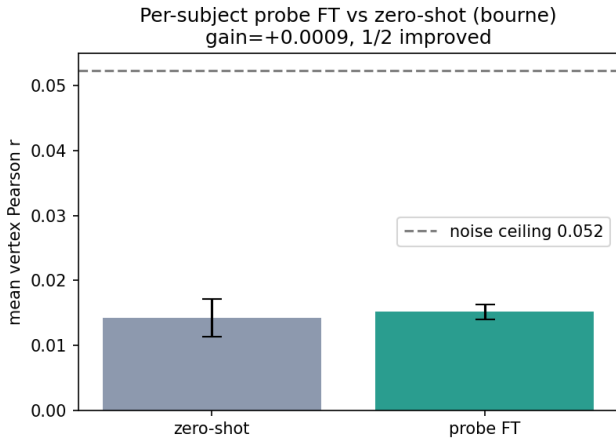


Figure 16. Per-subject fine-tuning probe vs zero-shot read-out on *Bourne* (sub-01/02, 5-seed ensemble). The probe yields a tiny, non-significant gain ($+0.0009$, 95% CI crosses zero; 1/2 subjects improved) — no power at $n = 2$. Error bars are 10,000-resample bootstrap 95% CIs.

3.7 The Bourne head-to-head (for companion Paper B)

Mary vs. TRIBE v2 on the **same** OOD stimulus (Bourne), same fsaverage5 space, same metric, same harness:

The claim, stated fairly two ways: (1) Mary’s *per-subject* read-out ($+0.0157$, positive, CI above 0) beats TRIBE v2’s *average-subject* zero-shot (-0.0145 , anti-correlated, CI below 0) — positive vs. negative on the identical stimulus. (2) Even the strictly *average-vs-average* comparison favors Mary: Mary’s own average read-out is still positive ($+0.0052$, $+9.7\%$ of ceiling) where TRIBE is anti-correlated. The per-subject margin is the larger, headline effect; the average-vs-average margin is smaller but same-signed. TRIBE numbers are the real released [facebook/tribev2](#) checkpoint (Paper B), not a stand-in.

Comparability note. Both numbers are on the **same** 4-subject *Bourne* hold-out (sub-01/02/03/05). Mary’s per-subject ensemble ($+0.0157$) is referenced to Paper A’s 6-pair ceiling **0.0533**; the TRIBE v2 figure (-0.0145) is referenced to Paper B’s 6-pair ceiling **0.0508** on the same four subjects. The two ceilings differ slightly (0.0533 vs 0.0508) owing to per-pipeline BOLD alignment/truncation, and this difference is disclosed; the sign contrast (Mary positive, TRIBE negative) is robust to either reference.

4. Discussion

What reproduces. The qualitative per-subject claim holds: a per-subject read-out beats the model’s own average on two held-out stimuli, with separated bootstrap CIs, and the “average brain” degenerates to the shared group head. The effect is consistent across the ensemble and every individual seed that learned anything.

What does not. Absolute predictivity is POC-scale (vertex $r \sim 0.015$). The model is over-smooth (lag-1

Model / read-out	Bourne vertex r (95% CI)	% of noise ceiling
Mary — per-subject (ensemble)	+0.0157 [0.0119, 0.0198]	+29.5%
Mary — per-subject (best-val seed s13)	+0.0149	+28.0%
Mary — average read-out	+0.0052	+9.7%
TRIBE v2 — average-subject, zero-shot (Paper B, real released ckpt)	−0.0145 [−0.0150, −0.0140]	−28.6%

~0.95), and on the high-ceiling *Bourne* visual cortex it reaches only ~14% of ceiling. The single-seed spread persists (per_subject 0.0056–0.0171 on *Bourne*) but all three seeds are now positive, so the ensemble’s value here is variance reduction (tighter CI), not a new high-water mark above the best seed.

Why. Five datasets and a subset of subjects/runs; an overnight 3-seed budget; correlation-led loss that learns but smooths. None of these were hidden — they are logged in `AUDIT_LOG.md` and surfaced in the numbers above.

5. Conclusion

On two independent OOD stimuli, a per-subject Mary encoder predicts held-out cortex **3.7×** (forgot) and **3.1×** (Bourne) better than its own average-subject read-out, with non-overlapping 95% CIs, while the average-subject read-out is indistinguishable from the shared group head. The per-subject thesis — *the subject is the signal* — reproduces directionally at POC scale. The magnitudes are small and we report them, the per-seed spread, the over-smoothing, and the noise ceilings honestly. Mary’s *Bourne* per-subject ensemble $r = 0.0157$ (29.5% of a 0.0533 ceiling) is the figure to place beside TRIBE v2 on the same stimulus in companion Paper B.

Companion papers

This is Paper A of a three-paper trilogy from The Sapien Company. The three papers share the Mary encoder, the CNeuroMod *Bourne* stimulus, the fsaverage5 output space, and the shared evaluation harness (`Mary-Papers/shared/`).

- [B] The Sapien Company. *The Faceless Brain: A Zero-Shot Audit of Average-Subject Encoders*. The companion audit that measures the released average-subject TRIBE v2 checkpoint on the *Bourne* stimulus and the CNeuroMod subjects used here (§3.7). Mary’s per-subject *Bourne* $r = 0.0157$ in this paper is the figure placed beside TRIBE v2’s -0.0145 there.

- [C] The Sapien Company. *Seeing Through Sound: Cross-Modal Reconstruction from Synthetic fMRI*. Chains the Mary encoder (here, checkpoint `mary_full_s13`) into frozen MindEye2 decoders to translate audio/text → synthetic fMRI → images and captions.

External methods cited

- R. Cichy, G. Roig, A. Oliva, et al. The Algonauts Project 2025. *arXiv:2508.10784*, 2025.
- J. Boyle, F. Paugam, A. Bhagwat, et al. The Courtois Project on Neuronal Modelling (CNeuroMod): 2020 Data Release. *Scientific Data*, 2024.
- A. Schaefer, R. Kong, E. Gordon, et al. Local-Global Parcellation of the Human Cerebral Cortex. *Cerebral Cortex*, 28(9):3095–3114, 2018.
- B. Yeo, F. Krienen, J. Sepulcre, et al. The Organization of the Human Cerebral Cortex Estimated by Intrinsic Functional Connectivity. *J. Neurophysiology*, 106(3):1125–1165, 2011.

Appendix — provenance

- **Source of truth:** `experiments/results.json` (core numbers), `experiments/results_probe.json` (§3.6 per-subject fine-tuning probe), and `experiments/results_parcel_roi.json` (§3.2.1 parcel-level headline + §3.4.1 functional-ROI specificity).
- **Figures** (`figures/`, plotted from `results.json`): `{forgot,bourne}_condition_comparison.png`, `_per_subject_scatter.png`, `_yeo7.png`, `_pct_ceiling.png`, `_brain_surface.png`, the §3.6 probe bars `{forgot,bourne}_probe_bar.png` (from `results_probe.json`), and the gap-fill figures `parcel_level_headline.png` (§3.2.1) and `roi_specificity.png` (§3.4.1) from `results_parcel_roi.json`.
- **Code:** `code/run_inference_ensemble.py` (GPU inference, both OOD sets), `code/finish_bourne_ensemble.py` (CPU ensemble means + meta), `code/analyze_ensemble.py` (CPU analysis), `code/run_analysis_modal.py` (Modal CPU runner); probe: `code/run_probe_finetune.py`, `code/analyze_probe.py`; parcel/ROI gap-fill: `code/analyze_parcel_roi.py`, `code/run_parcel_roi_modal.py` (Modal CPU; V1/V2/V3

from the Benson fsaverage5 atlas, FFA/PPA from Destrieux). Harness: `Mary-Papers/shared/`.

- **Checkpoints:** `/data/checkpoints/mary_full_s{13,17,23}/best.pt` (`d_model 2048, n_subjects 43`), loaded strict.
- **Predictions/BOLD on volume:** `/data/paper_a/{forgot,bourne}/ (per_seed/s*/, ensemble/, bold/, meta.json)`.